

2. Hair length in cats is controlled by a pair of alleles. The allele for short hair (**H**) is dominant to the allele for long hair (**h**).



(a) State what is meant by the terms:

(i) allele; [1]

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(ii) dominant; [1]

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(iii) recessive. [1]

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- (b) (i) A cat breeder crossed a homozygous short-haired cat with a long-haired cat. **Complete the Punnett square** to show the predicted genotypes of the offspring. Use the letters **H** and **h** for the alleles. [2]

Gametes		

- (ii) State the **phenotype** of the offspring in the F1 generation. [1]

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- (iii) **Complete the Punnett square** to show the possible genotypes of the offspring if two of the F1 offspring were crossed. [2]

Gametes		

- (iv) Using the results from (b)(iii), state how many kittens would be predicted to be short-haired in a litter of 8 kittens. [1]

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- (v) The cat breeder wanted to determine whether one of the short-haired cats was homozygous or heterozygous. She decided to breed the short-haired cat with a long-haired cat. Predict the phenotypes of the offspring you would expect if the short-haired cat was:
- I. Homozygous [1]

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- II. Heterozygous [1]

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